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Development of the national injury resource database (NIRD)

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ABSTRACT

Introduction: Lack of an accurate, publicly available database of burn/trauma resources creates challenges in providing burn care. In response to this gap, our group developed the National Injury Resource Database (NIRD), a comprehensive database of all US burn centers (BC) and trauma centers (TC) and their capabilities.

Methods: Lists of all national BC and TC were obtained from the American Burn Association (ABA), the American College of Surgeons, and every state department of health. Data was cross-checked and included BC/TC were linked with a 7-digit identification number using the American Hospital Association Quick Search guide. Each center's resources and verification status were validated with electronic or telephonic communications.

Results: The final database includes 135 BC and 617 TC, of which 18 are BC-only, 500 are TC-only, and 117 are combined BC/TC. ABA-verified BC (n = 76) are only found in Washington DC and 31 states, and 8 states have no BC. In the last 10 years, a net increase of 7 burn centers was found nationally. The ABA's online BC directory is outdated.

Conclusions: NIRD represents the only up-to-date, comprehensive listing of BC and TC in existence. It categorizes all currently operating BC and TC across myriad classifications of designation and capabilities.

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1. Introduction

The management of burned patients involves specific education, training, and resources. Burn centers provide specialized care through teams, processes, and facilities designed to optimize treatment and rehabilitation of these time-sensitive injuries.

The American Burn Association (ABA) validates burn center quality of care through a rigorous process that uses

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detailed guidelines to evaluate the resources and capabilities of each center that applies for verification. In addition to ABA verification, state agencies designate burn centers utilizing requirements set forth by local governments or other authorized entities [1]. Currently 2% of US hospitals have designated burn centers [2].

Despite advancements in burn center quality improvement programs and prehospital routing, more than two-thirds of severely burned patients are treated at non-verified burn centers [3]. It is unclear which of these non-verified burn centers meet state designation requirements. Further, only a number of states have formal designation requirements and the degree to which each resembles the ABA's national guidelines is unknown [4]. These discrepancies at the state level combined with the lack of an accurate, publicly available database to characterize burn resources create challenges in routing patients and providing burn care, especially in regard to disaster scenarios.

In an effort to define and improve access to care in our community; help state or local communities better assess their resources for injury care; and understand demographic, geographic, and economic factors affecting injury care, our team developed the National Injury Resource Database (NIRD). NIRD is a comprehensive database of all US burn centers recognized by the ABA and/or state health department and all level I and level II trauma centers verified by the American College of Surgeons Committee on Trauma (ACS/COT) and/or designated by each state.

2. Methods

To create NIRD, each state was queried alphabetically for the presence of U.S. burn centers and trauma centers. Burn and trauma center lists with verification/designation status were obtained through a combination of resources provided by the ABA [5,6], the ACS/COT [7], and all state departments of health (SDH). Location and contact information, bed count, and adult and pediatric capabilities were reported from these sources when available. The existing data was then cross-checked with institutional websites and official state agency documents with omission of redundant entries. The included burn and trauma centers were linked to existing hospitals with a 7-digit identification number using the American Hospital AssociationHospital Quick Search guide (AHA-ID) [8], and burn center totals and burn beds were matched using the 2021 Edition American Hospital Association Hospital Statistics [9].

The resulting unified database included all trauma centers that were verified as a level I or level II by the ACS/COT and/or designated by state health departments and all burn centers recognized by the ABA (both verified and non-verified) and/or recognized by the state with detailed information regarding adult and/or pediatric capabilities, hospital and burn bed count, and location and contact information.

Since burn and trauma centers can be both verified at the national level and designated at the state level, all centers included in the database can be categorized as: (1) centers verified by the ABA or ACS/COT only, (2) centers designated by the state only, and (3) centers both verified by their respective professional society and designated by the state. A fourth inclusion category exists for burn centers: (4) centers that are recognized by the ABA as non-verified burn centers and/or recognized by the state as "certified," "licensed," or "approved" burn centers/units in states that lack formal burn center designation requirements.

The presence of formal burn center designation was assessed by contacting all 51 SDH by phone or email. Burn centers that were only recognized (licensed, certified, approved) by state health departments that did not have a burn specialist on site were excluded. In addition, trauma centers recognized by the state as a level I or II but by the ACS/COT as a level III or IV were excluded. In cases where state trauma levels differed from ACS/COT records, the ACS/COT level was recorded. Further, if state adult/pediatric capabilities differed from that of the ACS/COT or ABA, the ACS COT- and/or the ABA-reported capabilities were recorded.

The number and percent of verified, designated, both verified and designated; and the number and percent of adult, pediatric, and combined adult and pediatric burn and trauma centers for inclusion were determined. Burn and trauma center numbers were totaled for each state across each database (ABA, AHA, and SDH) to examine the variability of publicly available databases currently characterizing burn center resources. The NIRD was then compared to the ABA's 2012 Burn Care Facilities to characterize the net effect of domestic burn center openings and closings over the last decade on US burn resources and, thus, access to care [10].

Finally, NIRD was combined with large, commercially available claims databases to create an IRB-approved, centralized database to evaluate the characteristics of both verified and non-verified and/or state designated burn centers. Analyses included hospital and burn intensive care unit (ICU) bed count, hospital revenue, Emergency Department (ED) visits, discharges, and urban versus rural geography. Categorical data was assessed using Fisher's exact and continuous variables were assessed with paired t-tests and confidence intervals.

3. Results

The final database included a total of 135 burn centers and 617 trauma centers. Of the 635 institutions housing these specialized centers, 18 were burn-only centers or burn centers located in institutions with level III or IV trauma centers that were excluded from our database. Five hundred were trauma-only centers, and 117 were co-located burn and trauma centers. Level I and level II ACS/COT-verified trauma centers (n = 436) were present in 46 states plus Washington DC (excluding Mississippi, Pennsylvania, Washington, and Wyoming), while state designated trauma centers (n = 616) were present in 49 states plus Washington DC (excluding Wyoming). Of these, only 1 (< 1%) was only ACS/COT-verified, 181 (29.3%) were only state designated, and 435 (70.5%) were both ACS/COT-verified and state designated (Fig. 1). With respect to burn centers, ABA-verified burn centers (n = 76) were found in 31 states plus DC. Eleven states (Alabama, Arkansas, Hawaii, Idaho, Kentucky, Maine, New Mexico, Oklahoma, South Carolina, Vermont, West Virginia) had no

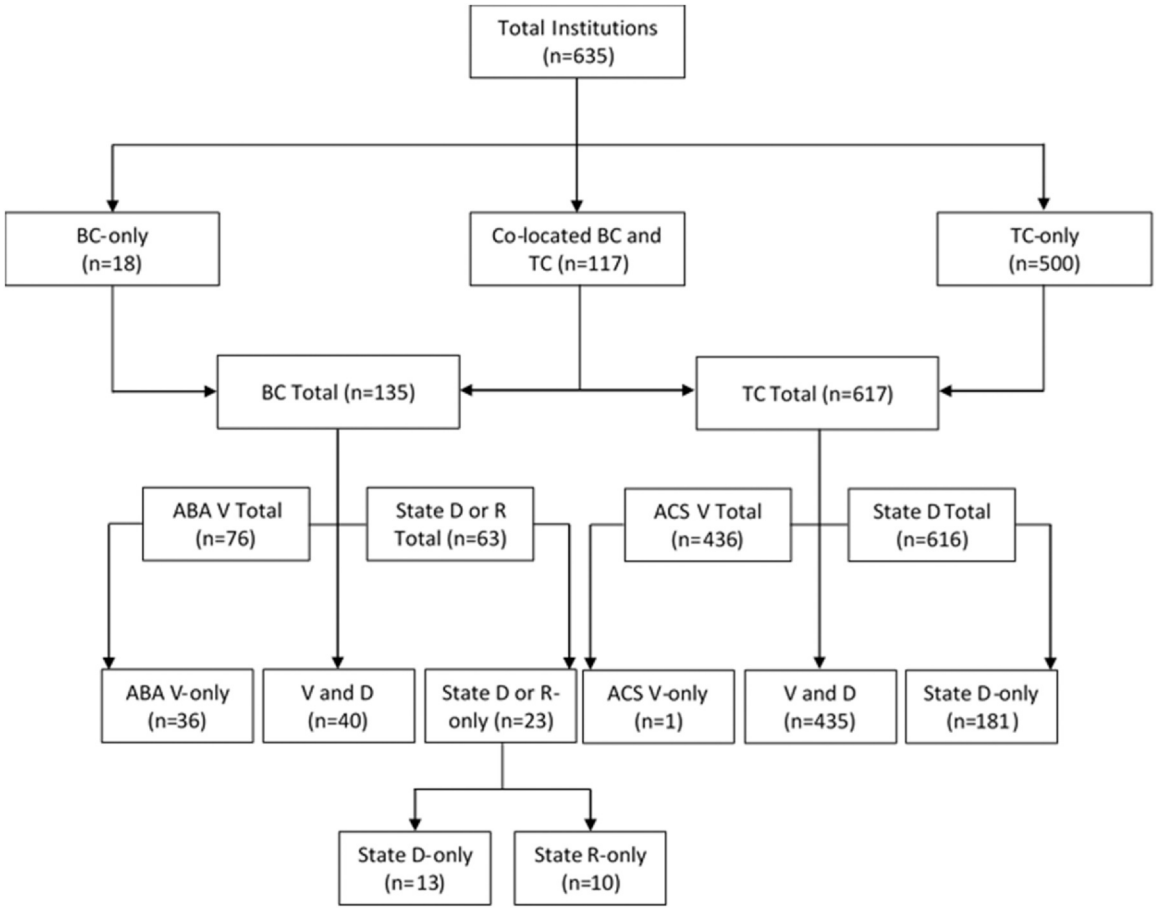

Fig. 1 – Summary of burn and trauma centers included in the National Injury Resource Database (NIRD) with verification and designation status numbers. BC= Burn Center, TC= Trauma Center, V= Verified, D= Designated, R= Recognized, ABA= American Burn Association, ACS= American College of Surgeons.

ABA-verified burn centers, and 8 states (Alaska, Delaware, Mississippi, Montana, North Dakota, New Hampshire, South Dakota, and Wyoming) had no burn centers at all.

While state trauma lists were readily available and easily accessible online through state department of health websites across Washington DC and all 50 states, state burn center lists were only available online for 15 states. Only 8 of these states (Connecticut, Georgia, Maryland, Nebraska, New Jersey, New York, Ohio, Virginia, and West Virginia) had formal burn center designation requirements, while 6 states (California, Washington DC, Florida, Louisiana, Massachusetts, and Michigan) lacked formal designation requirements, and instead, *recognized* their burn centers as licensed, certified, or approved burn centers or burn units (Table 1). We included a total of 29 designated and 34 recognized burn centers across these states. Officials in the remaining states claimed no official state designation system for burn centers in their states.

The proportion of adult and pediatric capabilities differed in burn and trauma centers. We found 472 (76.5%) adult-only trauma centers, 52 (8.4%) pediatric-only trauma centers, and 93 (15.1%) combined adult and pediatric trauma centers. These capabilities were difficult to obtain for burn centers, especially non-verified burn centers, as this information was not included in the ABA's publicly available resources. Through a combination of institution websites, state agency documents, and direct correspondence and phone conversation with burn center personnel, we identified 48 (35.6%) adult-only burn centers, 17 (12.6%) pediatric-only burn centers, and 70 (51.9%) adult and pediatric burn centers.

Comparing the NIRD to the ABA's 2012 Burn Care Facilities list, we found 12 burn centers have been removed from the ABA's burn center directory in the last decade after either closing (n = 10) or failing to maintain ABA recognition (n = 2). In addition, 18 burn centers have been added to the ABA's burn center directory, including those that have been transferred to another hospital or location, indicating a net increase of 7 burn centers since 2012. In the same time frame, changes in burn center verification status across the US has occurred, with 14 burn centers gaining ABA verification and 1 burn center losing verification.

Hospitals had a median of 492 staffed beds; 79,449 ER visits; and 18,437 discharges. Burn centers had a median of 11 burn ICU beds and 3285 hospital days per year. We found no significant differences in hospital size, ER volume, hospital beds, or burn ICU beds between verified and designated burn centers. Urban versus rural geography also did not differ between verified and non-verified and/or designated burn

Table 1 – Summary of states with designated and recognized burn centers including type of state recognition and number of designated or recognized burn centers by state. (BC= Burn Center).

Designated BC	Recognized BC	Type of State Recognition
Connecticut (n = 1)	California (n = 13)	Licensed burn centers
Georgia (n = 2)	District of Columbia (n = 2)	Recognized burn centers
Maryland (n = 2)	Florida (n = 6)	Licensed burn units
Nebraska (n = 1)	Louisiana (n = 4)	Certified burn centers
New Jersey (n = 1)	Massachusetts (n = 3)	Licensed burn centers
New York (n = 10)	Michigan (n = 6)	State burn centers
Ohio (n = 8)		
Virginia (n = 3)		
West Virginia (n = 1)		

Table 2 – Median values of hospital and burn ICU beds, burn days, ER visits, discharges, and revenues for verified and designated burn centers. (ICU= Intensive Care Unit; ER= Emergency Room).

.	Median Verified	Median Designated	*p* value (95% Confidence Interval)
Staffed Hospital Beds	462 (N = 65)	363 (N = 53)	0.12 (−25 to 199)
Burn ICU Beds	14 (N = 43)	10 (N = 25)	0.11 (−1 to 9)
Burn Days	3171 (N = 43)	2185 (N = 25)	0.08 (−151 to 2263)
ER Visits	79,642 (N = 64)	78,410 (N = 52)	0.26 (−8009 to 28,529)
Discharges	22,212 (N = 64)	15,639 (N = 55)	**0.02** (1202 to 13,281)
Total Revenue	$994 M (N = 64)	$652 M (N = 52)	**0.01** ($102 M to $756 M)
Medicare Revenue	$171 M (N = 57)	$83 M (N = 51)	**0.002** ($35 M to $158 M)
Inpatient Revenue	$2.1B (N = 64)	$1.3B (N = 52)	**0.01** ($2.2 M to $1.7B)

centers (p = 0.54). All categories of revenue were associated with burn center verification (Table 2).

4. Discussion

From March to April of 2022, our team cross-referenced multiple datasets (the ABA Burn Center Map and Burn Center List, the ACS/COT Hospital and Facilities List, the AHA 2021 Annual Survey data, and DC and all fifty state departments of health) to identify the verification and designation status, adult and pediatric capabilities, total bed counts, and location and contact information of 635 institutions, including 135 burn centers and 617 trauma centers. This information was compiled to create our combined injury database with the goal of improving injury care through the identification and analysis of available databases to discern practice patterns, develop benchmarking strategies, reduce delays in treatment, and improve access to injury care. To our knowledge, NIRD is the first comprehensive injury database of all professional society-verified and/or state-recognized burn centers and level I and level II trauma centers for adults and pediatrics in the United States. We plan to update the database semi-annually.

Efforts to characterize burn center trends were hindered by the absence of consensus defining burn center quality of care at the state level and substantial variation across publicly available databases characterizing burn center resources. While the ABA sets the standard as the only national-level program that verifies burn center quality of care, we found variability in their online resources in 18 burn centers across 15 states when comparing their Burn Center Directory to their Burn Center Regional Map, suggesting that their resources are outdated. This is compounded as the total number of verified burn centers varied across the ABA's sources with 76 verified burn centers per their online resources [5,6], 74 verified burn centers per an ABA employee via email correspondence, and 72 verified burn centers per the ABA's Executive Director in a recent podium presentation [11]. In addition to the inconsistencies in the ABA's resources, we found differences in the total number of burn centers reported across databases (ABA, AHA, SDH).

Previous studies have noted a significant decrease in the number of burn centers and available burn beds over previous decades [12–14]. However, the last 10 years have shown an upwards trend in US burn center numbers. Despite this, burn center capacity and capability, (defined as adequate beds and resources, and the presence of sufficiently experienced staff and providers, respectively) remains limited [15]. With only 250 dedicated burn surgeons in the US and burn centers operating at or over capacity on a daily basis, specialized care is scarce, and burn centers are limited to no more than five large burn admissions in the event of a disaster [13,16–18]. In addition to rising burn center numbers, we have also identified an increase in the number of verified burn centers over the last decade. Still, a large proportion of burn centers remain non-verified by the ABA, and thus, their quality of care is unknown. Given that only between 72 and 76 ABA verified burn centers are present and only 15 states with established designation requirements, a large number of burn centers operate without quality assurance. The lack of standardisation makes it difficult to determine non-verified burn center capabilities and to know where to route burns, especially in states without a verified burn center when transfers must be made across state lines [19].

Although the ABA has developed referral criteria to help ensure that severe burn injuries are treated at specialized burn centers [20], over half of burn care still occurs at non-

verified burn centers [21]. Previous studies have emphasized the variation in per-population access to verified burn center care, prompting the consideration of self-designated, non-verified centers as potential candidates for improving access [21]. Comparing characteristics of verified to non-verified or self-designated burn centers, we found no difference in hospital size, location (urban versus rural), ED volume, hospital beds, or burn ICU beds. Significant differences were only identified in the rate of discharge and revenues. This could be attributed to services, quality, or reimbursement.

These results suggest that the non-verified or self-designated burn centers included in our database may already be able to fulfill verification requirements set by the ABA. Thus, more state health departments should support the development of regional burn care and seek national-level verification by the ABA, or at least implement state regulations that mirror the ABA's guidelines. The implementation of these suggestions in trauma care has proven that standardized systems for designation and verification at the state and national level are effective in producing desired outcomes [22,23]. Additional research is needed to better understand these significant differences in verified and non-verified/self-designated burn centers, and further investigation is of use to determine how these compare in terms of patient outcomes.

The lack of management and tracking of burn resources, uneven distribution of burn centers, and the absence of standardized burn care requirements create challenges in burn care, especially in the event of a domestic or international conflict. Currently no commercially available data sets are available to support a detailed understanding of injury care in the US, and coordination and analysis of multiple datasets remains a substantial task of integrating technical and clinical expertise. An accurate, widely available injury database could help alleviate some of these issues. We believe that NIRD can serve as a foundation for the construction of publicly available real-time tools to aid in routing burn patients and improving transfer operations, increasing the accessibility of information, and improving access to care.

CRediT authorship contribution statement

This work has not been published elsewhere, nor is it in consideration at any other journals. All authors have done the work to merit co-authorship by making substantial contributions to all of the following: (1) the conception and design of the study, or acquisition of data, or analysis and interpretation of data, (2) drafting the article or revising it critically for important intellectual content, and (3) final approval of the version to be submitted. This manuscript will be presented as a poster at the 2023 annual meeting of the American Burn Association.

Declaration of Competing Interest

Mr. Phillips is the CEO of BData, Inc. Dr. Phelan serves as a consultant with Avita Medical and SpectralMD. Dr. Carter is a stockholder for Permeaderm, Inc. and SpectralMD, and serves as Chief Medical Consultant for SpectralMD. He also serves as a consultant for Avita Medical and Polynovo Ltd. where, in lieu of compensation, all proceeds are donated to a local burn charity to support burn education, survivor programs, outreach, and research. Dr. Hickerson serves as the Medical Monitor for SpectralMD, the Chief Medical Officer for Access Pro Medical, the Chief Medical Officer for SiOxMed, a consultant for Avita, and a consultant for PolyNovo.

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